

Omer Mujahid

Associate Researcher in AI for Healthcare · University of Girona, Spain

+34 632 671 443

omerm.pk@gmail.com

linkedin.com/in/omer-mujahid

ORCID: 0000-0001-9694-9621

EXPERIENCE

- Mar 2025 – Present* **Associate Researcher** · [University of Girona](#)
- Lead researcher on the Personal Digital Twin project (funded by Generalitat de Catalunya).
 - Driving formation of a university spinoff company in AI for diabetes.
 - Grant writing and supervision of PhD students.
- Sept 2023 – Feb 2025* **Postdoctoral Researcher** · [University of Girona](#)
- Advanced the MICELAB T1D physiological simulator using generative deep learning.
 - Co-led spinoff formation activities, grant applications, and PhD student mentorship.
- May – Dec 2021* **Research Fellow** · [Regional Academy on the United Nations \(RAUN\)](#)
- Contact tracing for COVID-19 in Ukraine — case study in Chernivtsi region (WHO Ukraine).
- Sept 2018 – Aug 2020* **Bachelor's Programme Advisor** · [CECOS University of IT & Emerging Sciences](#)
- Student guidance on academic regulations, final-year project selection, and career counselling.
 - Regular student welfare meetings covering health, financial, and emotional support.
- June 2018 – Aug 2020* **Postgraduate Studies Advisor** · [CECOS University of IT & Emerging Sciences](#)
- Supervised postgraduate research, assigned supervisors, streamlined thesis review processes.
 - Organised and coordinated thesis defence committees.
- Sept 2013 – Aug 2020* **Lab Engineer** · [CECOS University of IT & Emerging Sciences](#)
- Teaching simulation and design of signals & systems, digital filter design and analysis.
- Sept 2013 – Sept 2015* **Counsellor** · [CECOS IEEE Student Branch](#)
- Mentored students in IEEE activities; organised and facilitated branch events.
- Jan 2012 – Sept 2013* **Treasurer** · [CECOS IEEE Student Branch](#)
- Managed workshop revenue and event expenditures.

EDUCATION

- Sept 2020 – Sept 2023* **PhD in Technology** · [University of Girona](#)
- Thesis: Data-driven models for type 1 diabetes using generative deep learning · Cum Laude*
- 2014 – 2017* **MSc Electrical Engineering** · [CECOS University](#)
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2009 – 2013

BSc Electrical Engineering · CECOS University

Final-year project: Face recognition using principal component analysis · CGPA 3.10/4.00

TECHNICAL EXPERTISE

Generative Deep Learning	Digital & Virtual Twins	Physiological Modelling
Machine Learning	Retrieval-Augmented Generation	Reconfigurable Computing
FPGA-based Systems	AI for Healthcare	Grant Writing

JOURNAL PUBLICATIONS

- **Omer Mujahid**, Zahid Ullah, Hassan Mahmood, Abdul Hafeez. "Fast Pattern Recognition through an LBP driven CAM on FPGA". IEEE Access, 2018.
- Hassan Mahmood, Zahid Ullah, **Omer Mujahid**, et al. "Beyond the Limits of Typical Strategies: Resources Efficient FPGA-based TCAM". IEEE Embedded Systems Letters, 2018.
- **Omer Mujahid**, Zahid Ullah. "High Speed Partial Pattern Classification System using a CAM-based LBP Histogram on FPGA". IEEE Embedded Systems Letters, 2019.
- **Omer Mujahid**, Ivan Contreras, Josep Vehi. "Machine Learning Techniques for Hypoglycemia Detection: Trends and Challenges". Sensors, 2021.
- Muhammad Ibrahim, **Omer Mujahid**, et al. "An FPGA-Based Accelerated Mutation Detection System for the Tumor Suppressor Gene". IEEE Access, 2021.
- Josep Noguer, Ivan Contreras, **Omer Mujahid**, et al. "Generation of Individualized Synthetic Data for Augmentation of T1D Data Sets Using Deep Learning Models". Sensors, 2022.
- **Omer Mujahid**, Ivan Contreras, Aleix Beneyto, Ignacio Conget, Marga Giménez, Josep Vehi. "Conditional Synthesis of Blood Glucose Profiles for T1D Patients Using Deep Generative Models". Mathematics, 2022.
- Fahad Malook, **Omer Mujahid**, et al. "On Enhancing the Performance of IEEE 802.11ah by Employing a Dynamic Raw Approach in IoT Networks". Wireless Personal Communications, 2023.
- **Omer Mujahid**, Ivan Contreras, Aleix Beneyto, Josep Vehi. "Generative Deep Learning for the Development of a Type 1 Diabetes Simulator". Communications Medicine, 2024.
- Muhammad Bilal, Zahid Ullah, **Omer Mujahid**, Tama Fouzder. "Fast Linde-Buzo-Gray (FLBG) Algorithm for Image Compression through Rescaling Using Bilinear Interpolation". Journal of Imaging, 2024.
- Amina Sundas, Ivan Contreras, **Omer Mujahid**, et al. "The Effects of Environmental Factors on General Human Health: A Scoping Review". Healthcare, 2024.
- Josep Vehi, **Omer Mujahid**, Aleix Beneyto, Ivan Contreras. "Generative artificial intelligence in diabetes healthcare". iScience, 2025.
- Oriol Bustos, **Omer Mujahid**, et al. "Including Aerobic Exercise into Data-Based Virtual Twins for Glycemic Simulation". Journal of Diabetes Science and Technology, 2025.
- Chengli Liu, **Omer Mujahid**, et al. "Sex differences in glycemic outcomes: a systematic review and meta-analysis of diabetes treatments". BMJ Open Diabetes Research & Care, 2026.

CONFERENCE PAPERS & PRESENTATIONS

- **Omer Mujahid**, et al. "Design exploration of LH-CAM with updating mechanism". IJCCI 2019.
- Najib Ur Rehman, **Omer Mujahid**, et al. "Low Power Pre-Comparison Configuration Strategy for a Logic-based Binary CAM on FPGA". INTELLECT 2019.

- Najib Ur Rehman, **Omer Mujahid**, et al. "Power Efficient FPGA-based TCAM Architecture by Using Segmented Matchline Strategy". AECT 2019.
- Ali Mujtaba Durrani, **Omer Mujahid**, et al. "Micro Hydro Power Plant using Sewage Water of Hayatabad Peshawar". ICET 2019.
- **Omer Mujahid**, Ivan Contreras, Josep Vehi. "Generation of Realistic Virtual Diabetes Patients Using A Pix2pix GAN". Oral – Diabetes Technology Meeting 2021; J. Diabetes Science and Technology, 2022.
- Josep Noguer, **Omer Mujahid**, et al. "Improvement of Nocturnal Hypoglycemia Models by Using Synthetic Data from GANs". Poster – ATTD 2022.
- Oriol Bustos, Júlia Soler, **Omer Mujahid**, Josep Vehí. "Predictive Modeling of Exercise-Induced Glycemic Outcomes Using Generative Deep Learning". Oral – Diabetes Technology Meeting 2024; J. Diabetes Science and Technology, 2025.
- Josep Noguer, Ivan Contreras, **Omer Mujahid**, Josep Vehi. "Using conditional generative models to estimate exogenous glucose appearance rates in T1D". Poster – ATTD 2025.
- Elisa Pellizzari, **Omer Mujahid**, et al. "Enhancing Physiological Plausibility of GAN-Generated Blood Glucose with Monotonicity Constraints". MetroXRaine 2025.
- Oriol Bustos, **Omer Mujahid**, et al. "Personalized synthetic data generation for T1D patients using transfer learning". Poster – ATTD 2026.
- Marc Font, Oriol Bustos, **Omer Mujahid**, et al. "Implementation of physiology-informed loss in deep generative models for glycemic outcome predictions". Poster – ATTD 2026.
- Chengli Liu, **Omer Mujahid**, et al. "Sex differences in glycemic outcomes: systematic review and meta-analysis". Poster – ATTD 2026.

INVITED TALKS & WORKSHOPS

- "Enhancing predictive models thanks to synthetic data" — EMBC 2022.
- "Data-based digital twins: From twin cohorts to digital twins" — EMBC 2025 (Mini Symposium on Digital Twins for Diabetes).
- "Introductory Talk on AI in Diabetes Healthcare" — Design Thinking Workshop, Càtedra DEXCOM-UdG, 2025.
- "AI & Diabetes Management: An Introduction" — International Summer School on AI for Diabetes Management, UdG–Dexcom Chair, September 2025.
- "T1D Simulator and Digital Twins for Personalized Care" — International Summer School on AI for Diabetes Management, September 2025.
- "Artificial Intelligence in Diabetes" — AI and Nutrigenetics of Diabetes Mellitus Workshop, Institute of Nutrigenetics, October 2025.
- "Artificial Intelligence in Precision Nutrition" — International Conference on Precision Nutrition (Nutriome 2026), February 2026.
- "Building data-based digital twins in diabetes with deep generative models" — Oral presentation, ATTD 2026, Barcelona.

BOOK CHAPTERS

- Josep Vehi, **Omer Mujahid**, Ivan Contreras. "AI and Diabetes". Artificial Intelligence in Medicine, pp. 701–709. Springer, 2022. ISBN 978-3-030-64573-1.
- Josep Vehi, **Omer Mujahid**, Ivan Contreras. "Artificial intelligence and machine learning for diabetes decision support". Advanced Bioscience and Biosystems for Detection and Management of Diabetes, pp. 259–272. Springer, 2022. ISBN 978-3-030-99728-1.

GRANTS AWARDED

- "Vehicle number plate recognition" — National Grassroots ICT Research Initiative, 2017–18.
- "Diseased plant detection through a drone-mounted camera" — National Grassroots ICT Research Initiative, 2019.

- "Personal Digital Twin: A virtual human twin for therapy adjustment in diabetes" — Indústria del Coneixement 2024, Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR), Generalitat de Catalunya.

AWARDS & SCHOLARSHIPS

- Best Lab Engineer Award, 2015 — awarded by the President of CECOS University, Peshawar.
- FI Predoctoral Fellowship — Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR), Generalitat de Catalunya.

ADMINISTRATIVE RESPONSIBILITIES

- Coordinator, Summer School on AI in Diabetes — University of Girona.
- Postgraduate Studies Advisor — CECOS University.
- Councillor, IEEE Student Branch — CECOS University.
- Member, programme team for Self-Assessment Reports (SAR) for MS and PhD programmes.
- Member, examination panel for Final Year Project assessment.
- Organiser and programme committee member, IEEE ICET 2015, Peshawar.
- Industry–Academia Linkage Head, Open House 2019, CECOS University.
- Presented updated MS Electrical Engineering curriculum at the Board of Studies, November 2019.
- Convener, committee for the design of the Postgraduate Student Handbook.

UNDERGRADUATE PROJECTS SUPERVISED

- Developing a GAN-Based Blood Glucose T1D Outcome Prediction Model for Clinical Use (2024/25).
- Programació d'un simulador de dades de pacients amb diabetis per aplicacions en bioinformàtica (2020/21).
- A smart ECG platform using Deep Learning (2019/20).
- Internet packet classification using a CAM-based Hamming Distance Calculator on FPGA (2019/20).
- Neural Networks implementation on FPGA (2019/20).
- A novel distance calculation method for image classification on FPGA using TCAM (2018/19).
- Diseased plant detection through a drone-mounted camera (2018/19).
- Deep learning for image recognition through MATLAB (2017/18).
- Fast Pattern Recognition through CAM (2016/17).
- Exoskeleton design (2016/17).
- Industrial Automation through PLCs (2016/17).
- Multiple projects 2015/16: Dress Measurement App, My Team Society App, Automatic Car Control, Advanced Security System, Object-following robot (OpenCV).
- UHF/VHF Jammer (2014/15).